

TP53: Introduction and Role in Cancer Mutation Analysis

1. Gene Overview

TP53 is a tumor suppressor gene located at chromosome 17p13.1. One study in the reviewed literature reports the gene spans 19,149 base pairs across 11 exons. It encodes p53, a 393-amino-acid DNA-binding transcription factor that regulates the expression of numerous downstream genes.

2. Functional Role

p53 functions as a central mediator of the cellular stress response and a key anti-cancer safeguard (Donehower et al., 2019; Marei et al., 2021; Aubrey et al., 2016). Upon activation, it can arrest the cell cycle, promote DNA repair, induce senescence, or trigger apoptosis when damage is irreparable.

3. Relevance to Cancer

TP53 is among the most frequently altered genes in human cancer, with mutations reported in approximately 50% of cases (Wang et al., 2022; Cole et al., 2017; Olivier et al., 2009, and others). Loss of TP53 function is associated with increased chromosomal instability, altered transcriptional programs, and reduced survival in certain cancer types (Donehower et al., 2019; Li et al., 2018).

Notably, TP53 pathway inactivation is not always driven by mutation of the gene itself. A subset of tumors retain wild-type TP53 but exhibit functional p53 loss through dysregulation of upstream or downstream regulators such as MDM2, MDM4, PPM1D, or USP28 — a phenomenon referred to as phenocopying.

4. Considerations for Mutation Analysis

Second-hit events. TP53 status alone is insufficient to characterize pathway disruption; the mutation must be interpreted in context. In TCGA data, over 91% of TP53-mutant cancers showed a second-hit event affecting the remaining allele — via deletion, additional mutation, or copy-neutral loss of heterozygosity (Donehower et al., 2019). This indicates a tendency toward biallelic, rather than monoallelic, pathway inactivation (Donehower et al., 2019; Olivier et al., 2009).

Variant-specific effects. TP53 mutations are not functionally uniform. Loss-of-function effects are well established, dominant-negative effects are supported for many variants, and gain-of-function effects are reported in specific contexts but remain more contested (Kennedy et al., 2022; Liu et al., 2024).

Beyond hotspot codons. Over 20% of TP53 mutations occur outside the DNA-binding domain, and clinically relevant alterations also arise in intronic regions and gene

rearrangements. Analysis restricted to classic hotspot codons risks missing these events (Hainaut et al., 2016; Bouaoun et al., 2016).

5. Significance

- Loss of p53 function removes a critical barrier to uncontrolled proliferation, providing a selective advantage to cancer cells (Boutelle et al., 2021; Giacomelli et al., 2018).
- Most TP53 mutations are missense variants concentrated in the DNA-binding domain, frequently abolishing transcriptional activity (Donehower et al., 2019; Wang et al., 2022; Olivier et al., 2009).
- Mutation patterns can additionally reflect mutagenic exposure history and carry prognostic significance in specific tumor types (Donehower et al., 2019; Olivier et al., 2009; Hainaut et al., 2016).

6. Key Supporting Studies

Study	Focus	Key Finding
Olivier et al., 2009	Clinical overview	TP53 mutations are frequent and informative for prognosis and etiology
Fito-Lopez et al., 2022	Functional loss without mutation	Some tumors phenocopy TP53 loss without a detectable mutation
Donehower et al., 2019	TCGA pan-cancer analysis	Second-allele loss is highly prevalent in TP53-mutant cancers

Summary

TP53 encodes p53, a transcription factor central to cellular stress response and tumor suppression. Disrupted in roughly half of human cancers, its clinical interpretation requires more than mutation detection alone: evaluation of second-hit events, mutation-specific functional effects, and non-hotspot alterations is necessary to accurately assess p53 pathway status in a given tumor.

TP53 Mutation Analysis Across TCGA Studies — Study Report

1. Data Source

Mutation data for TP53 was obtained from cBioPortal, covering samples from 32 TCGA Pan-Cancer Atlas studies. Each sample is labeled either as wild-type (WT) or as carrying a TP53 mutation (mutant), with the mutant category including missense, nonsense, frameshift, and splice-site alterations as reported by cBioPortal.

2. Method

Figure 1. Samples were grouped by study ID, and for each study the number of WT versus mutant samples was counted. These counts were plotted as a stacked bar chart, with each bar representing one TCGA study and the bar height split between WT (blue) and mutant (orange) sample counts.

Figure 2. Using the same WT/mutant counts as Figure 1, the mutant count for each study was converted to a percentage of that study's total sample size (mutation frequency). These percentages were plotted as a single bar per study.

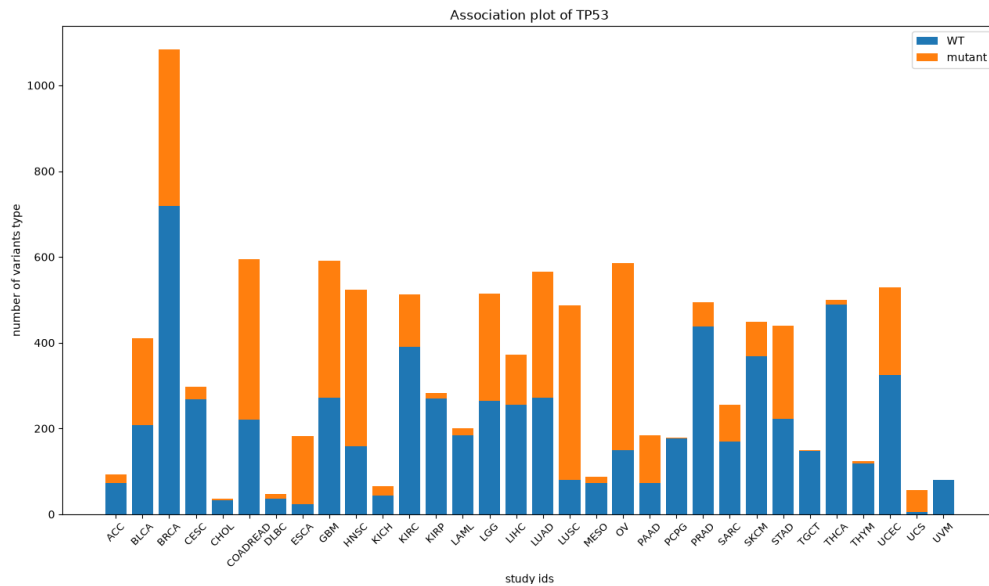
Figure 3. The mutant samples were further broken down by the specific amino acid position affected. Each distinct mutation position was counted across all studies combined, and the resulting frequencies were plotted as a bar chart sorted from most to least frequent, to identify the most commonly mutated positions.

Figure 4. The three most frequently mutated positions identified in Figure 3 (175, 248, 273) were selected, and for each TCGA study the number of mutations occurring at each of these three positions was counted. These counts were plotted as a heatmap, with studies on one axis and the three positions on the other.

Figure 5. The crystal structure of the p53 DNA-binding domain bound to DNA (PDB ID: 1TUP) was downloaded and visualized in ChimeraX. The three hotspot positions identified in Figures 3 and 4 (175, 248, 273) were manually located and labeled in red stick representation across all three protein chains in the structure.

Note: This is a beginner-level exploratory analysis throughout. The grouping, counting, and visualization methods used for Figures 1–4 are simple, and the structural labeling in Figure 5 was done manually. The observations in this report are based only on what was directly seen in the data, plots, and structure image — they may not account for confounding factors such as sample size differences, cohort composition, clinical variables, or structural details not captured by visual inspection alone.

3. Figure 1



Association plot of TP53: number of WT vs. mutant samples across TCGA studies.

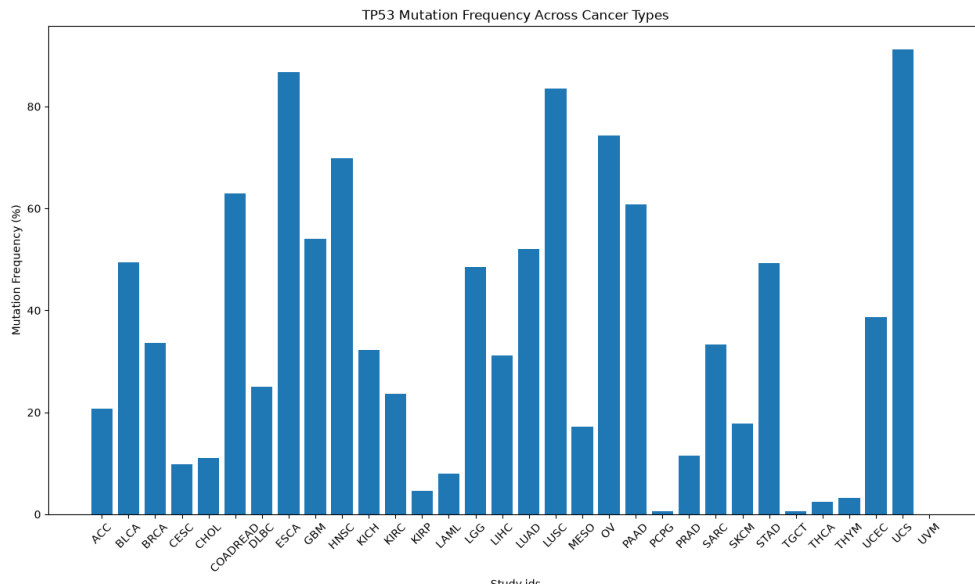
4. Observations

Overall dataset. Across all 10,967 samples in the dataset, 6,604 were WT and 4,363 carried a TP53 mutation, giving an overall mutation frequency of approximately 39.8%.

Largest cohorts. BRCA (breast cancer) had the largest sample size in the dataset (1,084 samples), with about 34% classified as mutant. GBM, COADREAD, LUAD, HNSC, and OV also had relatively large cohorts (over 550 samples each), each showing a substantial mutant fraction.

General pattern seen in the plot. Cancer types associated with squamous histology or gynecological/reproductive tissues (UCS, ESCA, LUSC, OV, HNSC) tend to show higher TP53 mutant proportions in this dataset, while endocrine and germ-cell-associated cancers (UVM, PCPG, TGCT, THCA, THYM) tend to show the lowest. This is only an observation from this specific plot and dataset, and no statistical test was performed to confirm significance.

5. Figure 2



TP53 Mutation Frequency Across Cancer Types: mutation frequency (%) per TCGA study, derived from the same WT/mutant counts as Figure 1.

6. Observations (Figure 2)

This figure presents the same underlying WT/mutant counts as Figure 1, but expressed as a mutation frequency percentage per study rather than raw sample counts. It is kept as a separate figure since it highlights relative frequency rather than absolute sample numbers.

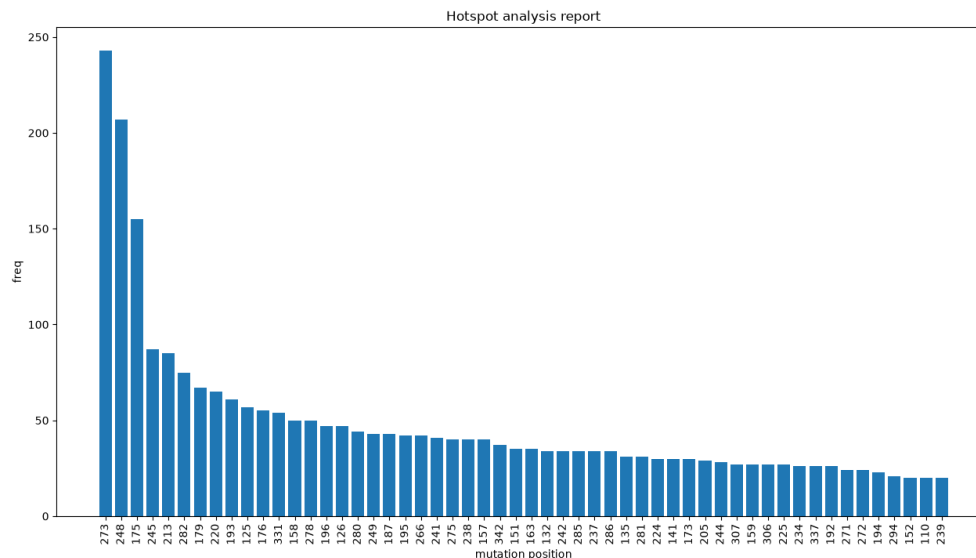
Highest mutation frequency. UCS (uterine carcinosarcoma) showed the highest proportion of mutant samples at roughly 91%, followed by ESCA (esophageal carcinoma, ~87%), LUSC (lung squamous cell carcinoma, ~84%), OV (ovarian cancer, ~74%), and HNSC (head and neck squamous cell carcinoma, ~70%). In the plot, these bars appear dominated by the orange (mutant) segment.

Lowest mutation frequency. UVM (uveal melanoma) showed no mutant samples in this dataset (0%). PCPG (pheochromocytoma and paraganglioma), TGCT (testicular germ cell tumors), THCA (thyroid carcinoma), and THYM (thymoma) also showed very low mutant proportions, each below 5%. These bars appear almost entirely blue (WT) in the plot.

Mid-range studies. A cluster of studies fall in the 30–55% range, including BLCA (~49%), BRCA (~34%), GBM (~54%), LGG (~48%), LUAD (~52%), STAD (~49%), and UCEC (~39%).

Comparison with Figure 1. Because Figure 2 removes the influence of cohort size and shows frequency directly, it makes the relative ranking of studies easier to read at a glance than the stacked counts in Figure 1 — for example, BRCA looked prominent in Figure 1 due to its large sample size, but Figure 2 shows its mutation frequency (~34%) is only moderate compared to smaller cohorts like UCS or ESCA.

7. Figure 3



Hotspot analysis report: frequency of TP53 mutations by amino acid position, across all studies combined, sorted from most to least frequent.

8. Observations (Figure 3)

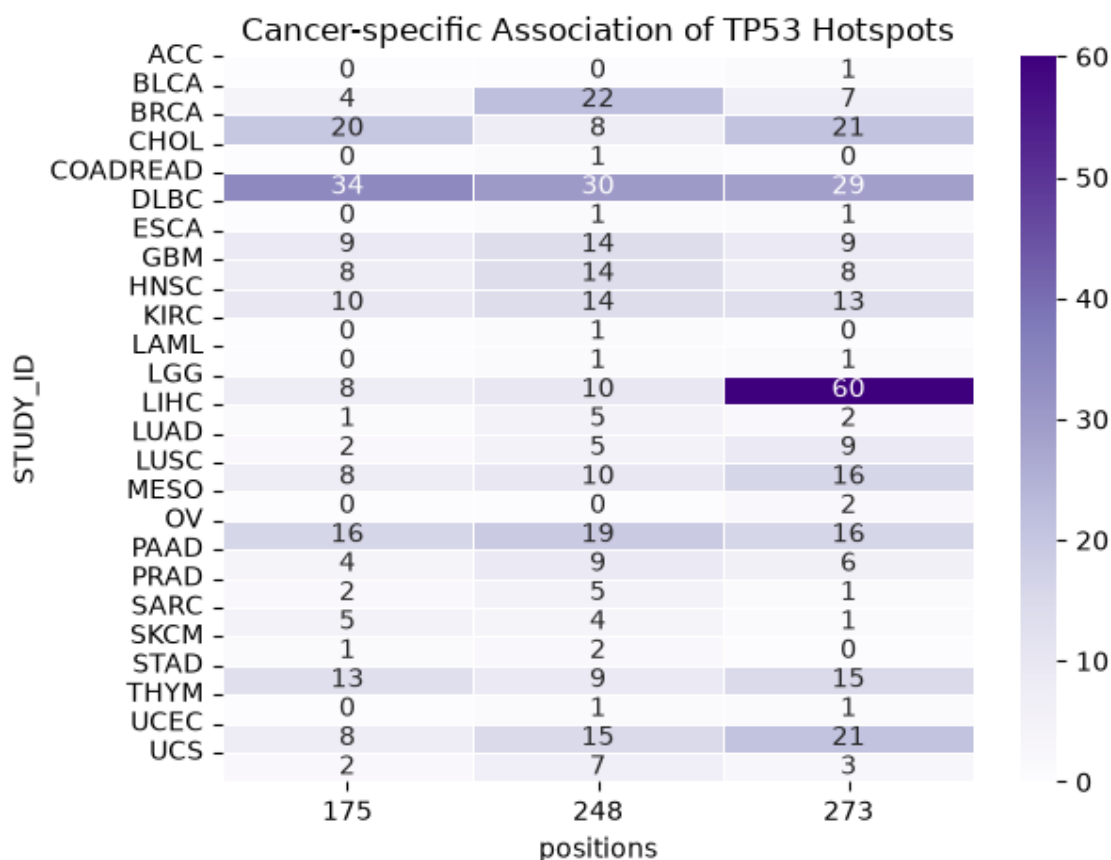
Dominant positions. Position 273 has the highest mutation frequency in the dataset (~243 occurrences), followed by position 248 (~207) and position 175 (~155). These three positions are clearly separated from the rest of the distribution by a visible drop after position 175.

Sharp drop-off. After the top three positions, frequency drops sharply to the 75–90 range for positions 245, 213, and 282, and continues to decline gradually from there.

Long tail. The remaining positions (e.g., 179, 220, 193, 125, 176, 331, and many others down to 239) form a long tail, each occurring roughly 20–65 times, with no single position standing out after the initial top three.

Interpretation. Positions 273, 248, and 175 are widely recognized in the literature as classic TP53 mutation hotspot codons, and this pattern is consistent with what the current dataset shows — a small number of positions accounting for a disproportionately large share of mutations, with the remainder spread thinly across many other positions.

9. Figure 4



Cancer-specific association of TP53 hotspots: mutation counts at positions 175, 248, and 273, broken down by TCGA study.

10. Observations (Figure 4)

Study-specific skew toward one position. LIHC showed a notably higher count of mutations at position 273 relative to positions 175 and 248 within this dataset.— this is the single highest count in the entire heatmap and is noticeably higher than any other study-position combination.

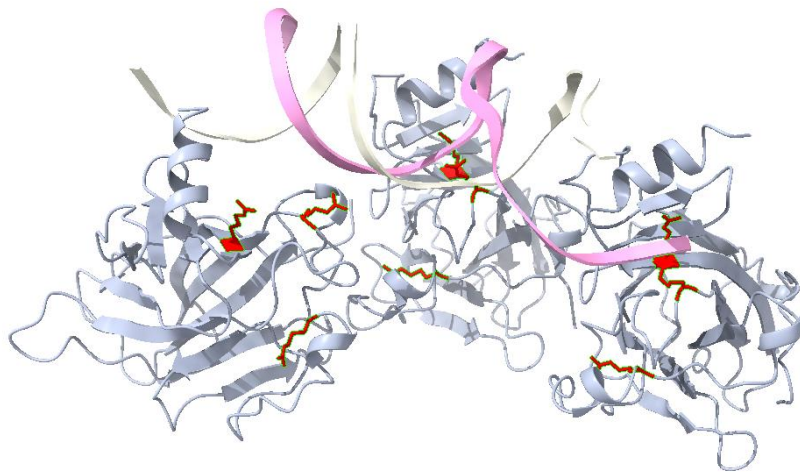
Studies with balanced distribution across all three positions. DLBC shows relatively high and fairly even counts across all three positions (34 at 175, 30 at 248, 29 at 273). BRCA (20, 22, 7) and OV (16, 19, 16) also show mutations spread across more than one position rather than being concentrated at a single one, though BRCA is comparatively lower at 273.

Studies with little to no hotspot mutations. ACC (0, 0, 1), COADREAD (0, 1, 0), LAML (0, 1, 0), and LGG (0, 1, 1) show almost no mutations at any of these three positions. PRAD (2, 5, 1) and SARC (5, 4, 1) are also comparatively low across the board.

No single position dominates uniformly. Aside from the LIHC/273 outlier, most studies do not show one position overwhelmingly dominating over the other two — counts across 175, 248, and 273 tend to be of a similar order of magnitude within a given study, suggesting

hotspot preference at these three positions may vary by cancer type rather than following one universal pattern.

11. Figure 5



p53 core domain–DNA complex (PDB 1TUP), rendered in ChimeraX, with the three hotspot positions (175, 248, 273) labeled in red stick representation across all three protein chains.

12. Observations (Figure 5)

Overall architecture. The structure shows three copies of the p53 DNA-binding domain (light blue/grey ribbons) arranged around a central DNA double helix (pink/tan ribbons), consistent with the known 1TUP co-crystal structure of the p53 core domain bound to its DNA response element.

Repeated hotspot labeling across chains. The three hotspot positions (175, 248, 273) are labeled in red on all three protein chains, giving nine labeled residues in total. The same three positions appear at structurally equivalent locations in each chain, which is expected since all three chains represent the same protein sequence.

Position relative to the DNA interface. Some of the labeled red residues are positioned close to the central DNA ribbon, near the top of the structure where the protein domains meet the double helix. Others, positioned lower and more toward the outer edges of each domain, sit further from the DNA and appear to lie within the body of the folded domain rather than at the direct DNA contact surface.

Symmetry of arrangement. The three chains are arranged in a roughly symmetric fashion flanking the DNA, with each chain's labeled residues mirrored in position relative to the other two chains.

Interpretation (tentative). The visual difference between hotspot residues sitting near the DNA interface versus those buried within the domain body may reflect two different ways a mutation at these positions could disrupt p53 function — by directly interfering with DNA contact, or by destabilizing the domain's fold. This is only an interpretation based on visual inspection of this single structure image, not a measured or verified structural analysis (e.g., no distance or contact-map data was calculated).

13. Limitations

- This analysis only counts WT vs. mutant status; it does not account for mutation type, zygosity, or co-occurring alterations.
- Sample sizes vary considerably across studies (from 36 in CHOL to 1,084 in BRCA), which may affect how reliable the mutant percentage is for smaller cohorts.
- No statistical testing (e.g., chi-square) has been applied to confirm whether differences between studies are significant.
- Figures 3 and 4 only look at three selected positions (175, 248, 273) and simple frequency counts; other hotspot positions, mutation subtypes at each position, and sample-size differences between studies were not adjusted for.
- The apparent skew in Figure 4 (e.g., LIHC at position 273) is based on raw counts within this dataset only and has not been checked against cohort size or verified with any statistical test.
- Figure 5's structural observations are based only on visual inspection of a single rendered image; no distance measurements, contact analysis, or independent verification of residue labeling was performed.
- This is an early-stage, self-directed exploration of the dataset carried out with simple counting, plotting, and visual inspection methods. All observations above reflect only what was directly seen in these five figures and this dataset — they are not validated findings and may not hold up or generalize in a real-world clinical or research setting.